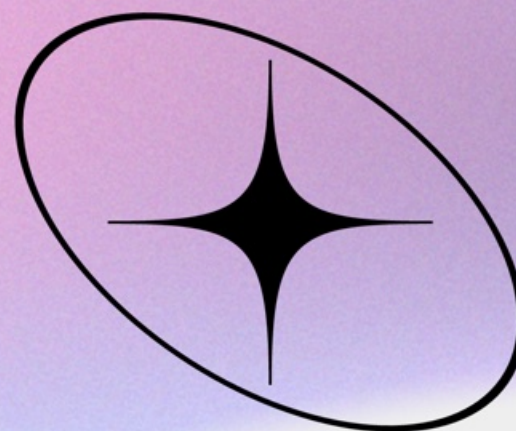


TER

MARS 2026

REPORT WEEK 1



EDNA & ROSEMARIE



TER - WEEK 1

```
=== SGEMM Benchmark (Single Core) ===
```

Size	OpenBLAS(GF/s)	Our SGEMM(GF/s)	Ratio
----	-----	-----	-----
256	157.98	53.18	0.34x
512	361.10	107.45	0.30x
1024	454.82	134.16	0.29x
2048	541.63	263.36	0.49x

OBSTACLE 1:

Scatter/Gather C Write-Back in the Micro-Kernel

→ the original micro-kernel used column-accumulators but C is row-major

SOLUTION:

→ switch to row-accumulator layout

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OBSTACLE 2:

Register-Limited 8×8 Kernel (Insufficient Compute Density)

→ 6 idle registers, wasted computing power

SOLUTION:

→ 6×16 Micro-Kernel (Matching OpenBLAS Design)

OBSTACLE 3:

Suboptimal Cache-Blocking Tile Sizes

→ the original tile sizes were MC=512, KC=256, NC=4096

SOLUTION:

→ MC must be a multiple of MR=6, best results with MC=120, KC=512, NC=4096

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OBSTACLE 4:

Thread Pool Churn and Serial B-Packing

→ overhead from creating/joining threads + one thread packs the entire B panel (up to 8 MiB) while 15 threads idle

SOLUTION:

→ single Parallel Region + Parallel B-Packing

OBSTACLE 5:

Insufficient Parallelism for Small Matrices

→ small matrices had very few parallel tasks, for $N=256$, 13 out of 16 threads sat idle during the compute phase

SOLUTION:

→ 2D Parallel Decomposition: instead of parallelizing only the ic loop, we flatten the (ic, jr) space into a single parallel loop

FINAL RESULTS

1 THREAD

Size	OpenBLAS(GF/s)	Our SGEMM(GF/s)	Ratio
----	-----	-----	-----
256	67.85	64.72	0.95x
512	71.65	67.29	0.94x
1024	73.50	67.88	0.92x
2048	74.62	65.64	0.88x

16 THREADS

Size	OpenBLAS(GF/s)	Our SGEMM(GF/s)	Ratio
----	-----	-----	-----
256	293.68	271.96	0.93x
512	427.49	417.71	0.98x
1024	484.05	453.45	0.94x
2048	528.08	547.50	1.04x

AND THEN...

WE RAN OUT OF CREDITS

